

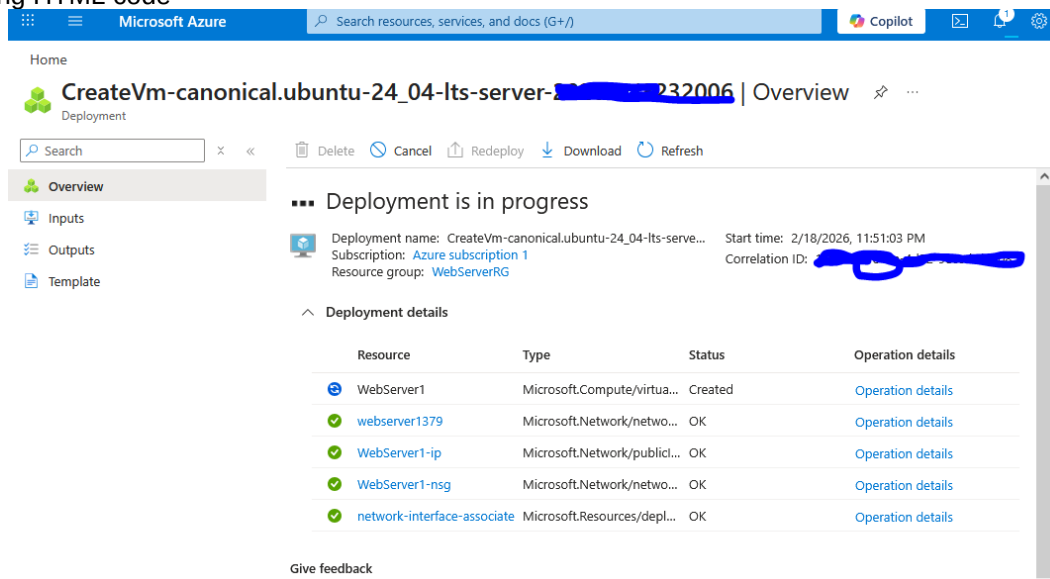
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COMP4358(1) - Wireless Networking
February 19, 2026

Creating a Virtual Machine and Running a Web Server in Microsoft Azure

Creating the Azure Virtual Machine

I logged into Azure Portal and created a new Ubuntu Server 24.04 LTS VM named WebServer1 in the East US region. I generated an SSH key pair (WebServer1_key) for secure remote access.

Fig 1: Custom web page displaying personalized content with course information and nano editor showing HTML code



The custom page shows "Welcome to Craig Ssemakula's Web Server!" with Azure hosting details, Ubuntu + Apache2 info, and COMP4358(1) course information. The nano editor displays the HTML and CSS code used.

Fig 2: Apache2 default page displaying 'It works!' confirming successful web server installation

Microsoft Azure

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Overview

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Essentials

Resource group (...)	: WebServerRG	SKU	: Standard
Location (move)	: East US	Tier	: Regional
Subscription (move)	: Azure subscription 1	IP address	: 13.91.217.102
Subscription ID	: 50c5c471-8b31-4612-8000-000000000000	DNS name	: -
See more		Associated to	: webserver1379
		Virtual machine	: WebServer1

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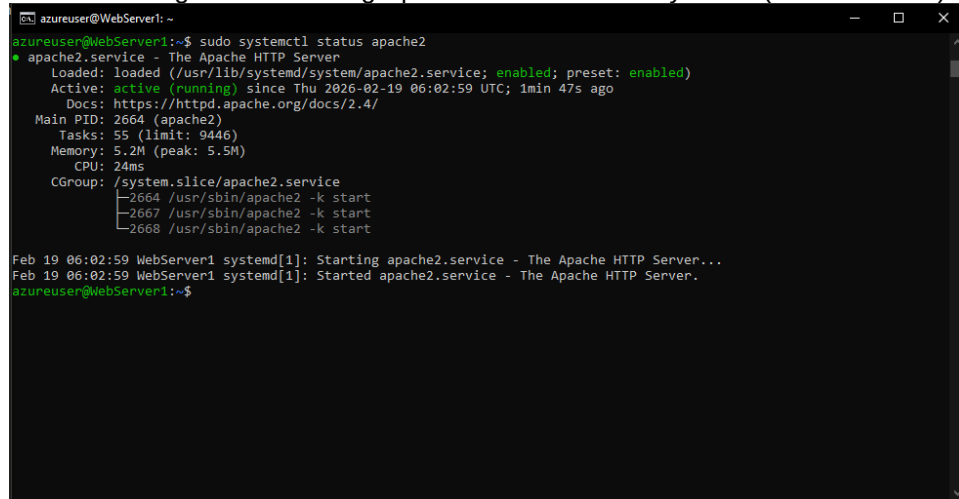
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The default Apache2 Ubuntu page confirmed successful installation and network configuration.

Installing and Configuring Apache Web Server

I connected via SSH using: `ssh -i WebServer1_key.pem azureuser@104.211.29.192`. Then I installed Apache2 and configured it to start automatically on boot.

Fig 3: UFW firewall configuration showing Apache allowed from anywhere (IPv4 and IPv6)



```
azureuser@WebServer1:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-02-19 06:02:59 UTC; 1min 47s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2664 (apache2)
    Tasks: 55 (limit: 9446)
   Memory: 5.2M (peak: 5.5M)
      CPU: 24ms
   CGroup: /system.slice/apache2.service
           └─2664 /usr/sbin/apache2 -k start
             └─2667 /usr/sbin/apache2 -k start
               └─2668 /usr/sbin/apache2 -k start

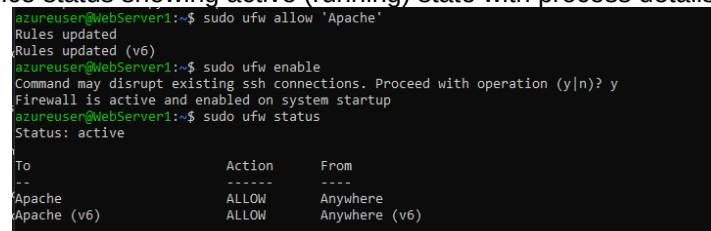
Feb 19 06:02:59 WebServer1 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Feb 19 06:02:59 WebServer1 systemd[1]: Started apache2.service - The Apache HTTP Server.
azureuser@WebServer1:~$
```

The firewall shows Apache allowed on both IPv4 and IPv6, permitting web traffic from any location.

Configuring Network Security

I configured UFW to allow HTTP traffic on port 80 by enabling the firewall and adding Apache to allowed applications.

Fig 4: Apache2 service status showing active (running) state with process details



```
azureuser@WebServer1:~$ sudo ufw allow 'Apache'
Rules updated
Rules updated (v6)
azureuser@WebServer1:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y/n)? y
Firewall is active and enabled on system startup
azureuser@WebServer1:~$ sudo ufw status
Status: active

To Action From
--
Apache ALLOW Anywhere
Apache (v6) ALLOW Anywhere (v6)
```

Apache2 showed active and running with process ID 2664, using 5.2MB memory. Started on February 19, 2026 at 6:02 AM.

Testing the Web Server

I tested the server by accessing 104.211.29.192 in a web browser. The Apache2 default page appeared, confirming everything worked correctly.

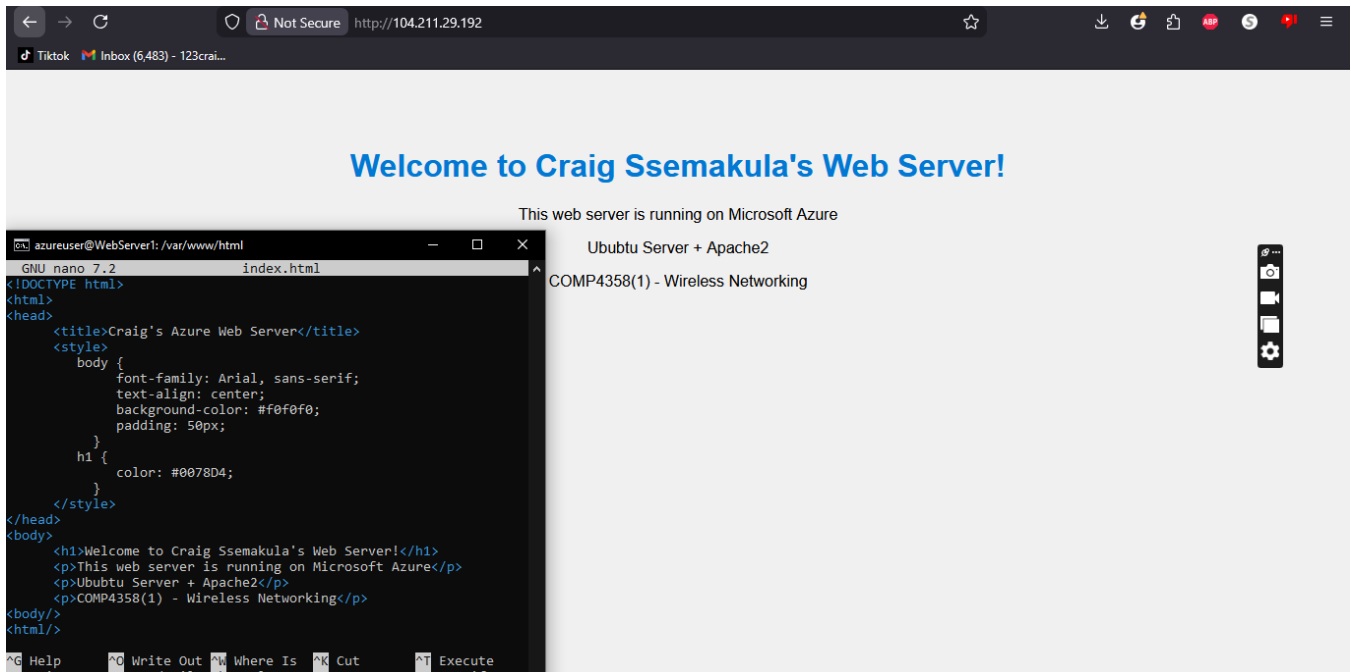
Fig 5: Public IP address resource (WebServer1-ip) showing the assigned IP: 104.211.29.192 in East US region



Customizing the Web Page

I edited /var/www/html/index.html using nano to create a personalized page with my name, course info, and server details.

Fig 6: Azure deployment in progress showing VM creation with all resources (WebServer1, network interface, public IP, network security group)



The deployment created the VM, network interface (webserver1379), public IP (WebServer1-ip), and network security group (WebServer1-nsg). All resources deployed successfully in about 2-3 minutes.

What I Learned

This lab taught me practical cloud infrastructure skills:

Cloud Computing: How to provision and manage VMs in Azure.

Linux Administration: SSH access, package management with apt, and service control with systemctl.

Network Security: Configuring firewall rules to control traffic and manage ports.

Web Server Setup: Installing Apache2, understanding web directories, and editing HTML files.

IP Management: How Azure assigns public IPs for internet access.

Conclusion

Successfully deploying a web server on Azure covered the complete workflow of cloud hosting. The lab demonstrated how cloud platforms make it easy to deploy globally accessible web applications.